PART 1 : PHYSICS

SECTION-I (i)

1) If the third harmonic vibration in an open air pipe equals the fifth harmonic of vibration in a closed air pipe, the possible values of the length of air column in the closed pipe and open pipe, respectively, are (neglect end corrections) :-

(A) 240 cm, 120 cm

(B) 60 cm, 72 cm

(C) 200 cm, 120 cm

(D) 180 cm, 150 cm

2) Two balls having mass 2 kg and 3 kg are approaching each other with velocities 3m/s and 2 m/s respectively on the horizontal frictionless surface. They undergo a head on elastic collision. Find out the maximum potential energy of deformation.

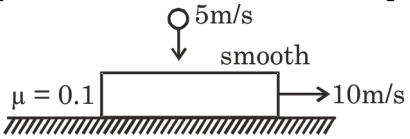
(A) zero

(B) 12.5 J

(C) 15 J

(D) none of these

3) A ball of mass m collides elastically with a plank moving with speed of 10 m/s on a rough horizontal surface. The upper surface of the plank is smooth but the lower surface is rough with coefficient of friction, $\mu = 0.1$. The mass of the plank is also m. What is the speed of the plank just after the collision ? The speed of ball is 5 m/s :-



(A) 9 m/s

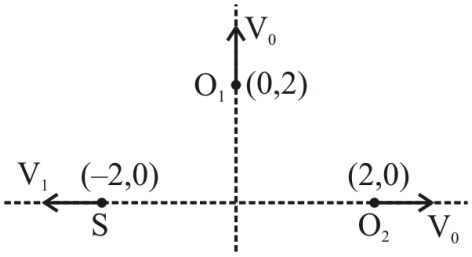
(B) 8 m/s

(C) 9.5 m/s

(D) 10 m/s

4) At t = 0 a sound source S and observers O_1 & O_2 are placed as shown. S is moving with speed V_1 and O_1 and O_2 are moving with speed V_0. At any later instant, let f_1 and f_2 represent apparent frequencies of sound observed by O_1 and O_2 respectively. The ratio f_1/f_2 is (Note : V_1 and V_0 are less

than speed of sound in this medium.)



(A) zero

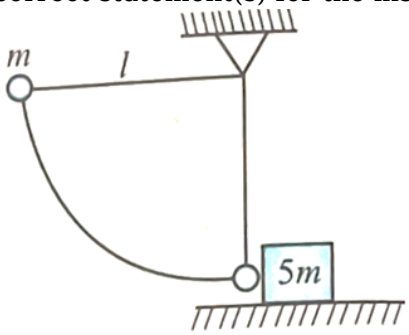
(B) 1

(C) between 0 and 1

(D) greater than 1

SECTION-I (ii)

1) A pendulum bob of mass m connected to the end of an ideal string of length ℓ is released from rest from horizontal position as shown in figure. At the lowest point, the bob makes an elastic collision with a stationary block of mass 5m, which is kept on a frictionless surface. Mark out the correct statement(s) for the instant just after the impact.



(A) Tension in the string is (17/9) mg

(B) Tension in the string is 3 mg

(C) The velocity of the block is $\sqrt{2g\ell/3}$

(D) The maximum height attained by the pendulum bob after impact is (measured from the lowest position) $4\ell/9$.

2) A triangular wave pulse on a taut string travels in positive x-direction with speed v. The tension in the string is F, and linear mass density of string is μ. At t = 0, the shape of pulse is given by

$$y(x,0) = \begin{cases} 0 & \text{if} \quad x < -L \\ \frac{h(L+x)}{L} & \text{for} \quad -L < x < 0 \\ \frac{h(L-x)}{L} & \text{for} \quad 0 < x < L \\ 0 & \text{for} \quad x > L \end{cases}$$

Choose the incorrect statement(s)

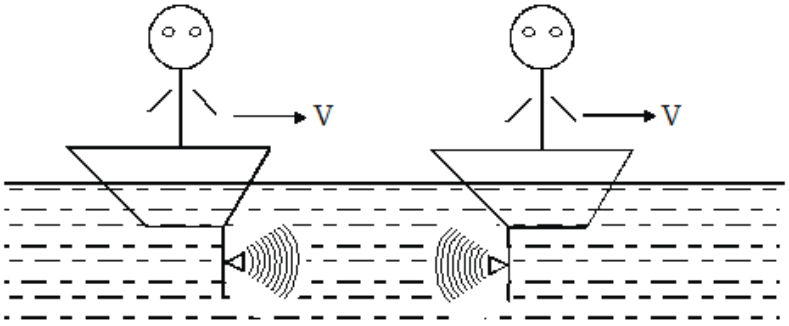
(A) Magnitude of instantaneous power is zero for –L < (x – vt) < 0

(B) Magnitude of instantaneous power is $Fv\left(\frac{h}{L}\right)^2$ for (x – vt) < –L

(C) Magnitude of instantaneous power is $Fv\left(\frac{h}{L}\right)^2$ for (x – vt) > L

(D) Magnitude of instantaneous power is $Fv\left(\frac{h}{L}\right)^2$ for 0 < (x – vt) < L

3) Two boats are rowing on a pond in the same direction and with the same speed V. Each boat sends, through the water (which is at rest), a signal to the other. The frequencies of the generated signals are the same.



(i) The time taken by the signals to travel from one boat to the other is different.
(ii) The wavelengths received by both the persons in the boats due to the other are different.
(iii) The beat frequencies received by both the persons are different.
(iv) The beat frequencies received by both the persons are same (which may include zero also)
Choose the INCORRECT option(s) :

(A) (i), (ii), (iii) are correct

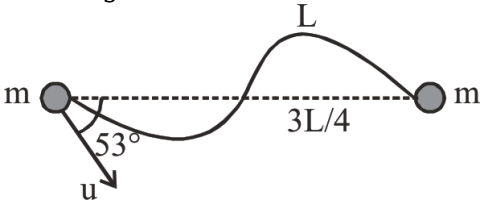
(B) (ii), (iii) are correct

(C) (iii) is correct

(D) (i), (ii), (iv) are correct

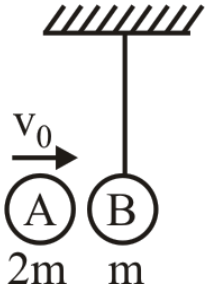
SECTION-II

1) In the figure shown, each tiny ball has mass m, and the string has length L. One of the balls is imparted a velocity u, in the position shown, in which the intial distance between the balls is $\frac{3L}{4}$. If the impulse of the tension in the string when it becomes taut is mα kg-ms^{-1}, then find the value of α.

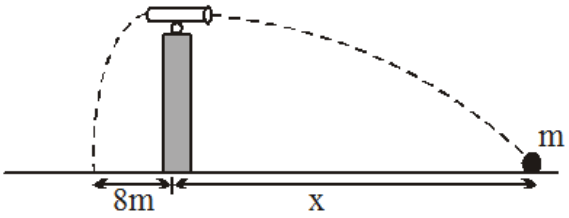


(Given : m = 2kg, u = 15 m/s)

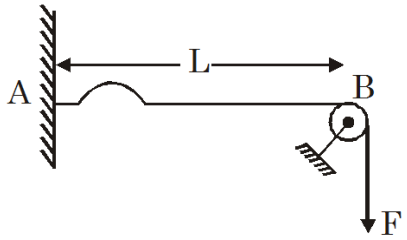
2) A sphere A of mass 2kg moving with velocity v_0 collide with bob of mass 1kg as shown in the figure. Value of tension at which string of pendulum break is 40N. If length of pendulum is 30cm, then find the maximum value of v^0 in m/s so that string does not break. Coefficient of restitution is $\frac{1}{2}$.



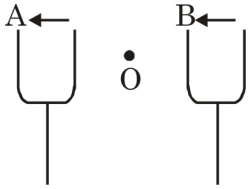
3) A cannon of mass M = 200 kg is positioned a top a narrow wall as shown. It fires a ball of mass m = 2kg horizontally across a plane. Unfortunately the gunners forgot to lock the frictionless wheels of the cannon and it immediately rolls backwards off the wall, landing a distance 8 m from the wall as shown. Neglecting air friction, at what distance x (in meters) from the base of the wall does the ball land ?



4) In the given figure, a string of linear mass density 2 gm/cm and length L = 2m is stretched by a force F = kt where k is constant and t is in sec. If at time t = 0, a pulse is generaged at the end A of the string and it will reach to end B in 1 sec. Find the value of k in S.I. unit.

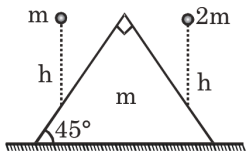
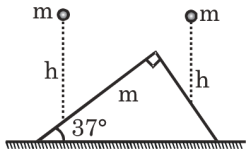
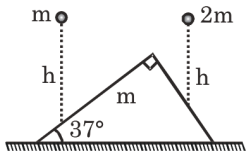
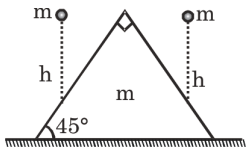


5) Two tuning forks A and B each of natural frequency 85 Hz move with velocity 10 m/s relative to stationary observe 'O'. Fork A moves away from the observer while the fork B moves towards him as shown in the figure. A wind with a speed 20 m/s is blowing at an angle 60° with the direction of motion of fork A. Find the beat frequency measured by the observer in Hz. [Take speed of sound in air as 340 m/s]



6) A stationary tuning fork is in resonance with an air column in a pipe. If the tuning fork is moved with a speed of 2 ms^{-1} in front of the open end of the pipe and parallel to it, the length of the pipe should be changed for the resonance to occur with the moving tuning fork. If the speed of sound in air is 320 ms^{-1}, the smallest value of the percentage change required in the length of the pipe is _____.-

1) In the given figures two balls are dropped from height h on a wedge which is placed on smooth surface as shown in figure. If collisions are perfectly inelastic and collisions are occur simultaneously. Then :-

List-I		List-II	
(P)		(1)	Net impulse in horizontal direction is towards right on the wedge.
(Q)		(2)	Net impulse in horizontal direction is towards left on the wedge.
(R)		(3)	Wedge will move towards right.
(S)		(4)	Wedge will move towards left.
		(5)	Wedge remains stationary.

(A) P → 2,4;Q → 2,4;R → 2,4;S → 5

(B) P → 1,2;Q → 2,3;R → 3,4;S → 5

(C) P → 2,4;Q → 2,4;R → 2,5;S → 5

(D) P → 3,4;Q → 4,5;R → 5;S → 2

2) A standing wave, formed in an organ pipe aligned along the x-axis, is given by
$S = (5 \times 10^{-6}\ m) \sin(4\pi x) \cos(1320\pi t)$
Where x is in m and t is in seconds.

List-I		List-II	
(P)	Magnitude of change in pressure is maximum	(1)	$x = \frac{1}{4}m$
(Q)	Magnitude of change in pressure is minimum	(2)	$x = \frac{1}{2}m$
(R)	Maximum amplitude of vibration	(3)	$x = 1m$
(S)	Minimum amplitude of vibration	(4)	$x = \frac{1}{8}m$
		(5)	$x = \frac{3}{8}m$

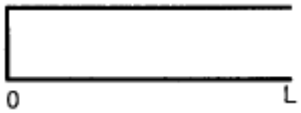
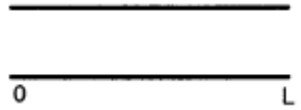
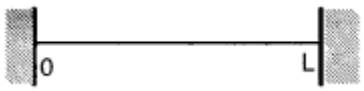
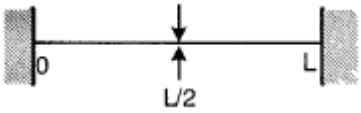
(A) P → 1,2,3;Q → 4,5;R → 4,5;S → 1,2,3

(B) P → 1,3;Q → 3,5;R → 2,3,4,5;S → 1,2,3,4,5

(C) P → 2,3;Q → 2,4;R → 2,3,5;S → 3,4,5

(D) P → 1,3,4;Q → 1,3;R → 2,3;S → 2,3,4,5

3) **List-I** shows four systems, each of the same length L, for producing standing waves. The lowest possible natural frequency of a system is called its fundamental frequency, whose wavelength is denoted as λ_f. Match each system with statements given in **List-II** describing the nature and wavelength of the standing waves.

List-I		**List-II**	
(I)	Pipe closed at one end 	(P)	Longitudinal waves
(II)	Pipe open at both ends 	(Q)	Transverse waves
(III)	Stretched wire clamped at both ends 	(R)	$\lambda_f = L$
(IV)	Stretched wire clamped at both ends and at mid-point 	(S)	$\lambda_f = 2L$
		(T)	$\lambda_f = 4L$

(A) I → P,T;II → P,S;III → Q,S;IV → Q,R

(B) I → P;II → R;III → P;IV → S

(C) I → P;II → T;III → P;IV → R

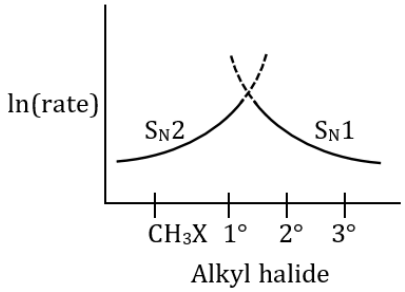
(D) I → P;II → R;III → P;IV → S

PART 2 : CHEMISTRY

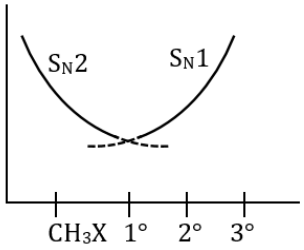
SECTION-I (i)

1) Which of the following curves correctly represents S_N1

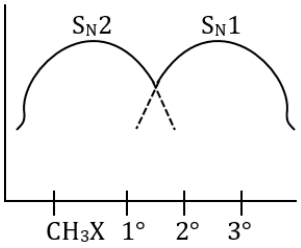
(A)



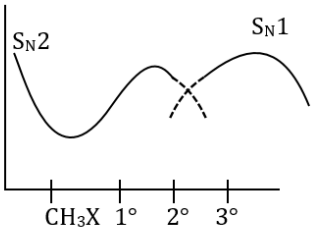
(B)



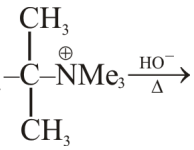
(C)

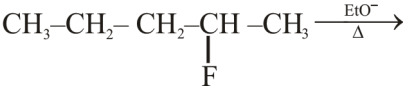


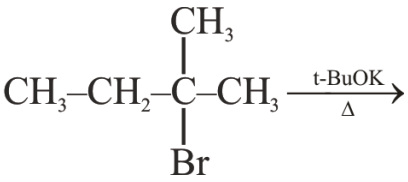
(D)

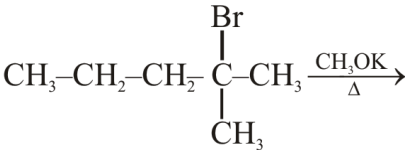


2) In which of the following reaction saytzeff alkene is major product ?

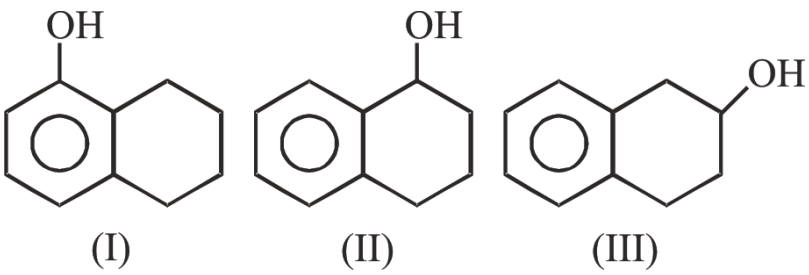
(A) 

(B) 

(C) 

(D) 

3) Compare the rate of acid catalyst dehydration:-



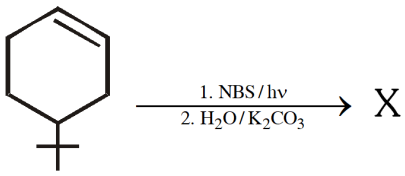
(A) I > II > III

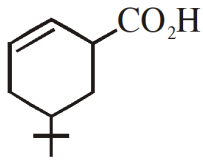
(B) II > III > I

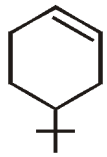
(C) III > II > I

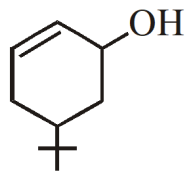
(D) II > I > III

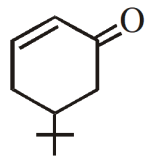
4) The product of the reaction given below is :



(A) 

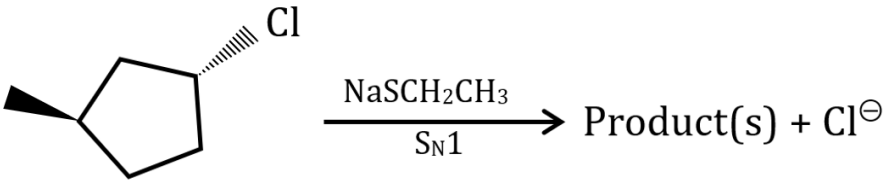
(B) 

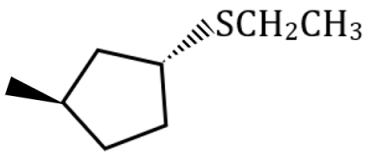
(C) 

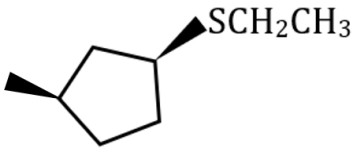
(D) 

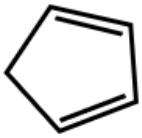
SECTION-I (ii)

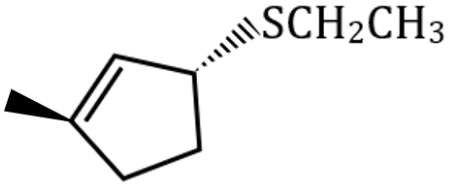
1) What could be the product(s) for the following S_N1 reaction?



(A) 

(B) 

(C) 

(D) 

2) Which of the following will give allyl halide?

(A) $H_2C = CH - CH_3 \xrightarrow[hv]{SO_2Cl_2}$

(B) $H_2C = CH - CH_3 \xrightarrow{Cl_2, 800K}$

(C) $H_2C = CH - CH_3 \xrightarrow{NBS}$

(D) $H_3C - CH = CH_2 + HBr \xrightarrow{H_2O_2}$

3) Aryl halides are less reactive towards nucleophilic substitution reaction as compared to alkyl halides due to

(A) The formation of less stable carbonium ion

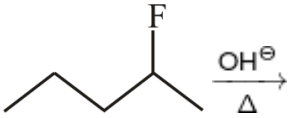
(B) Resonance stabilization

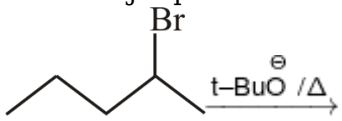
(C) The inductive effect

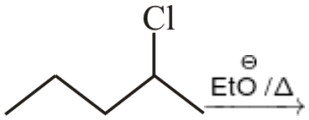
(D) sp^2 hybridised carbon attached to the halogen

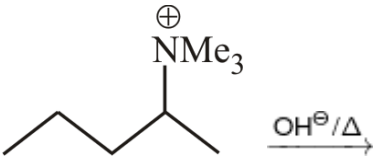
SECTION-II

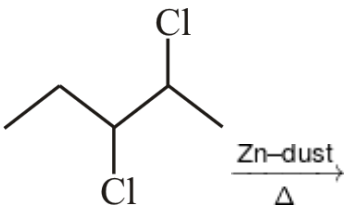
1) In which reaction Hoffmann alkene is the major product?

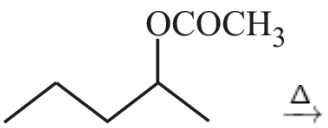
(i) 

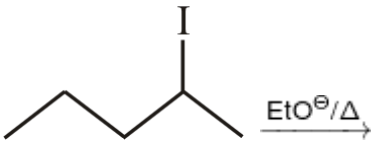
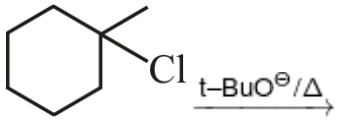
(ii) 

(iii) 

(iv) 

(v) 

(vi) 

(vii)  (viii) 

2) How many of the following substrates are more reactive than ethyl bromide for S_N2 reaction ?

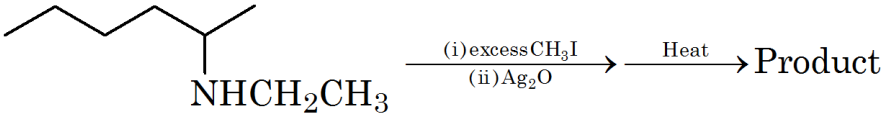
(a) CH_3-Br (b) O_2N-CH_2Br

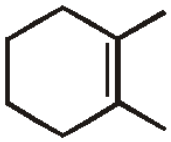
(c) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{C}}H-Br$ (d) $CH_2=CH-Br$

(e) $OHC-CH_2-Br$ (f) $CH_2=CH-CH_2-Br$

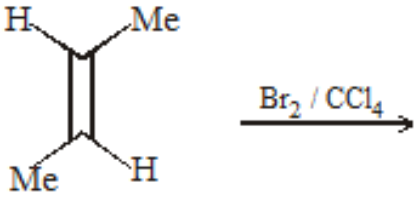
3) The total monochloro derivatives of C_4H_{10} are (only structural)

4) Predict the possible number of alkenes in the following reaction.



5)  $\xrightarrow{NBS}$ Monobromosubstituted products formed.

Ratio of total monobromo products formed to 2° monobromo products formed (include stereoisomers)

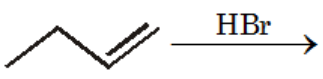
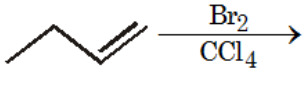
6) 

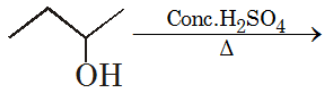
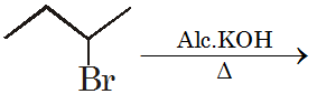
Product 'P'

How many stereo isomer of product "P" is/are possible?

SECTION-I (iii)

1)

List-I		List-II	
(P)		(1)	Unimolecular elemination
(Q)		(2)	Bimolecular elemination

(R)		(3)	Racemic mixture formed
(S)		(4)	Major product shows stereoisomerism
		(5)	Electrophilic addition

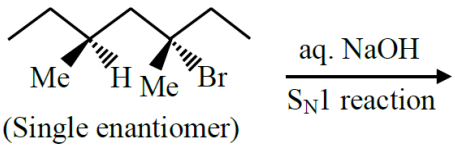
(A) P → 1,3,4,5;Q → 3,4,5;R → 1,4;S → 2,4

(B) P → 3,4,5;Q → 3,4,5;R → 1,4;S → 2,3,4

(C) P → 3,4,5;Q → 3,4,5;R → 1,4;S → 2,4

(D) P → 4,5;Q → 3,4,5;R → 1,4;S → 2,4

2) Match the reactions in List-I with the features of their products in List-II and choose the correct option.

List-I		List-II	
(P)	(-)-1-Bromo-2-ethylpentane (single enantiomer) $\xrightarrow[S_N2 \text{ reaction}]{\text{aq. NaOH}}$	(1)	Inversion of configuration
(Q)	(-)-2-Bromopentane (single enantiomer) $\xrightarrow[S_N2 \text{ reaction}]{\text{aq. NaOH}}$	(2)	Retention of configuration
(R)	(-)-3-Bromo-3-methylhexane (single enantiomer) $\xrightarrow[S_N1 \text{ reaction}]{\text{aq. NaOH}}$	(3)	Mixture of enantiomers
(S)	 (Single enantiomer)	(4)	Mixture of structural isomers
		(5)	Mixture of diastereomers

(A) P → 1;Q → 2;R → 5;S → 3

(B) P → 2;Q → 1;R → 3;S → 5

(C) P → 1;Q → 2;R → 5;S → 4

(D) P → 2;Q → 4;R → 3;S → 5

3) Match the entries on the left with most appropriate choice(s) given on the right.

Column-I		Column-II	
(A)	Dehalogenation of meso-2, 3-dibromobutane	(p)	S_N2
(B)	Dehydrohalogenation of 2-bromobutane using SN^2 path	(q)	Cis-2-butene
(C)	Hydrolysis of 2-bromo-3-methylbutane	(r)	S_N1
(D)	$CH_3CH = CHCH_2Cl + CN^- \rightarrow$ $CH_3\text{–}\underset{\displaystyle CN}{\underset{\vert}{CH}}\text{–}CH=CH_2$	(s)	Trans-2-butene
		(t)	Anti-elimination

(A) A → P;B → R;C → Q,T;D → S,T

(B) A → R;B → P;C → S,T;D → Q,T

(C) A → S,T;B → Q,T,S;C → R;D → R

(D) A → Q,T;B → P;C → R;D → S,T

PART 3 : MATHEMATICS

SECTION-I (i)

1) The integral $\int \frac{dx}{(x+4)^{\frac{8}{7}}(x-3)^{\frac{6}{7}}}$ is equal to :
(where C is a constant of integration)

(A) $\left(\frac{x-3}{x+4}\right)^{\frac{1}{7}} + C$

(B) $-\left(\frac{x-3}{x+4}\right)^{-\frac{1}{7}} + C$

(C) $\frac{1}{2}\left(\frac{x-3}{x+4}\right)^{\frac{3}{7}} + C$

(D) $-\frac{1}{13}\left(\frac{x-3}{x+4}\right)^{-\frac{13}{7}} + C$

2) If f(x) = $\int e^x \left(\tan^{-1}(x) + \frac{2x}{(1+x^2)^2}\right) dx$, f(0) = –1 then the value of f(1) is

(A) $e\left(\frac{\pi}{4} - \frac{1}{2}\right)$

(B) $e\left(\frac{\pi}{4} + \frac{1}{2}\right)$

(C) $e\left(\frac{\pi}{2} - \frac{1}{4}\right)$

(D) $e^{-1}\left(\frac{\pi}{4} - \frac{1}{2}\right)$

3) $\int_{-\pi/2}^{\pi/2} (\sin^2 x - \sin^4 x)$ (sin x – cos x) dx is equal to

(A) 4/15

(B) 0

(C) –4/15

(D) 2/15

4) Let $J = \int_0^{e-1} \frac{1}{x+1} \exp.\left(\frac{x^2+2x-1}{2}\right) dx$ and $K = \int_1^e x\, \ell n\, x\, \exp.\left(\frac{x^2-2}{2}\right) dx.$ The value of

(J + K) is equal to

(A) $\left(\sqrt{e}\right)^{e^2+1}$

(B) $\left(\sqrt{e}\right)^{e^2-1}$

(C) 0

(D) $\left(\sqrt{e}\right)^{e^2-2}$

SECTION-I (ii)

1) If $\int \frac{\left(\frac{1}{x}+\frac{1}{x^2}\right)(x-1)}{\left(\frac{1}{x^4}+\frac{1}{x^2}\right)\sqrt{\left(x^4-x^3+x^2\right)\left(x^4+x^3+x^2\right)}}dx = \sec^{-1}(f(x)) + C,$ then

(A) Maximum value of f(x) is –2 when $x < 0$

(B) Minimum value of f(x) is 2 when $x > 0$

(C) f(0) is not defined

(D) $f(e) < f(\pi)$

2)

If $I_m = \int_0^{m\pi} x\,|\sin x|\, e^{\cos 4x}dx$ and $J_n = \int_0^{n\pi} |\cos x|\, e^{\cos 4x}dx$, where $m, n \in I$ then

(Note : [k] denote GIF)

(A) $\left[\frac{I_1}{J_1}\right] = 0$

(B) $\left[\frac{I_2}{J_2}\right] = 3$

(C) $\left[\frac{I_2}{J_1}\right] = 0$

(D) $\left[\frac{I_3}{J_3}\right] = 4$

3) In which of the following situations, it is possible to have a triangle ABC?
(All symbols used have usual meaning in a triangle.)

(A) $(a + c - b)(a - c + b) = 4bc$

(B) $b^2 \sin 2C + c^2 \sin 2B = ab$

(C) $a = 3, b = 5, c = 7$ and $C = \frac{2\pi}{3}$

(D) $\cos\left(\frac{A-C}{2}\right) = \cos\left(\frac{A+C}{2}\right)$

SECTION-II

1) If $\int \frac{\sin x}{\sin x - \cos x} dx = G(x)$, then $\frac{24}{\pi}\left[G\left(\frac{\pi}{3}\right) - G\left(\frac{\pi}{6}\right)\right]$ is equal to

2) If f(x) = $\int \left(x^2 + 2x^4 + 3x^6\right)\left(1 + x^2 + x^4\right)^{\frac{1}{2}} dx$ and f(0) = 0, then value of $(f(1))^2$ is

3) Let $I_{(m,n)} = \int (\cos 2x)^m \left(1 - \tan^2 x\right)^n dx$, such that $4I_{(0,10)} + \frac{I_{(1,10)}}{5} = A \sin 2x \left(1 - \tan^2 x\right)^{10} + BI_{(0,9)} + C$ (where C is integration constant), then value of $\frac{B}{8A}$ is

4) For $a, b \in \mathbb{R}^+$, $\int_a^b \frac{e^{\frac{x}{a}} - e^{\frac{b}{x}}}{x} dx$ is

5) If the limit $\lim_{n\to\infty} n^2 \int_0^{1/n} x^{x+1} dx$ can be expressed in the form $\frac{m}{n}$, (where m and n are relatively prime positive integers), then the value of (m + n) is

6) In ΔABC, R_1, R_2 are circumradius of ΔABC and ΔOAB respectively, where O is circumcentre of ΔABC. If $1 - 2\sin\left(\frac{3A-B}{2}\right)\cos\frac{C}{2} = \cos C + 2\cos 2B$, $\angle A = \frac{5\pi}{12}$ and $\frac{R_1}{R_2} = k$, then value of $(k^2 - 2)$ is

SECTION-I (iii)

1) Match the List

	List-I		List-II
(P)	In a ΔABC, 2B = A + C and $b^2 = ac$. Then the value of $\frac{a^2(a+b+c)}{3abc}$ is equal to	(1)	8
(Q)	In any right angled triangle ABC, the value of $\frac{a^2+b^2+c^2}{R^2}$ is always equal to (where R is the circumradius of ΔABC)	(2)	1
(R)	In a ΔABC if a = 2, bc = 9, then the value of 2RΔ is equal to	(3)	5

(S)	In a ΔABC, a = 5, b = 3 and c = 7, then the value of 3 cos C + 7 cos B is equal to	(4)	9
		(5)	6

(A) P → 2;Q → 1;R → 4;S → 3

(B) P → 4;Q → 3;R → 5;S → 3

(C) P → 3;Q → 1;R → 4;S → 2

(D) P → 2;Q → 3;R → 5;S → 3

2) List-I contains 4 definite integrals and List-II contains the value desired. Match the entries.

	List-I	List-II	
(P)	$I_1 = \int_{-\pi/2}^{\pi/2} \cos^{-1}(\sin x)\,dx$ Let , then the value of $\frac{8I_1}{\pi^2}$ is	(1)	1
(Q)	$I_2 = \int_{0}^{\pi} \tan^{-1}(\cos x)\,dx$ Let , then the value of $I_2 + 4$ is	(2)	2
(R)	$I_3 = \int_{0}^{\pi/2} x^2 \sin 2x\,dx$ Let , then the value of $\frac{3}{4}(\pi^2 - 8I_3)$ is	(3)	3
(S)	$I_4 = \int_{\pi}^{2\pi} x^{[\sin x]}dx$ Let , then the value of e^{I_4} is coprime with where [*] denotes greatest integer function	(4)	4
		(5)	5

(A) P → 4;Q → 3;R → 1;S → 2

(B) P → 4;Q → 4;R → 3;S → 2

(C) P → 2;Q → 4;R → 3;S → 2

(D) P → 2;Q → 1;R → 3;S → 2

3)

	List-I		List-II
(P)	If $\int \frac{2^x}{\sqrt{1-4^x}}dx = k\sin^{-1}(f(x)) + C$, then k is greater than	(1)	1
(Q)	If $\int \frac{x^{5/2}}{(\sqrt{x})^7 + x^6}dx = a\ell n\left(\frac{x^k}{1+x^k}\right) + C$, then ak is less than	(2)	7
(R)	If $\int \frac{x^4+1}{x(x^2+1)}dx = \ell n\left\lvert\frac{x}{x^2+1}\right\rvert + \frac{x^k}{k} + C$, then k is less than	(3)	3

(S)	If $\int \frac{dx}{5+4\cos x} = k\tan^{-1}\left(m\tan\frac{x}{2}\right)+C$, then $\left(\frac{k}{m}\right)$ is greater than	(4)	4
		(5)	2

(where C is constant of integration)

(A) P → 1;Q → 3,4,5;R → 3,4;S → 1

(B) P → 1;Q → 1,2,3;R → 2,4;S → 1,3

(C) P → 1,3;Q → 3,4;R → 2,3;S → 1,4

(D) P → 1,3;Q → 3,4,5;R → 3,5;S → 1,2

ANSWER KEYS

PART 1 : PHYSICS

SECTION-I (i)

Q.	1	2	3	4
A.	B	C	A	D

SECTION-I (ii)

Q.	5	6	7
A.	A,C,D	A,B,C	A,B,C

SECTION-II

Q.	8	9	10	11	12	13
A.	6.00	3.00	800.00	1.80	5.00	0.62 to 0.63

SECTION-I (iii)

Q.	14	15	16
A.	A	A	A

PART 2 : CHEMISTRY

SECTION-I (i)

Q.	17	18	19	20
A.	B	D	B	C

SECTION-I (ii)

Q.	21	22	23
A.	A,B	A,B,C	B,D

SECTION-II

Q.	24	25	26	27	28	29
A.	5.00	4.00	4.00	4.00	3.50	3.00

SECTION-I (iii)

Q.	30	31	32
A.	C	B	C

PART 3 : MATHEMATICS

SECTION-I (i)

Q.	33	34	35	36
A.	**A**	**A**	**C**	**D**

SECTION-I (ii)

Q.	37	38	39
A.	**A,B,C,D**	**B,D**	**B,C**

SECTION-II

Q.	40	41	42	43	44	45
A.	**2.00**	**3.00**	**5.00**	**0.00**	**3.00**	**1.00**

SECTION-I (iii)

Q.	46	47	48
A.	**A**	**B**	**A**

SOLUTIONS

PART 1 : PHYSICS

1) $\frac{3}{2L}v = \frac{5}{4L'}v$

$\frac{L'}{L} = \frac{5}{6}$

2)

Max potential energy of deformation = max. K.E. loss

$= \frac{1}{2} \times 3 \times 2^2 + \frac{1}{2} \times 2 \times 3^2 = 15\ J$

3) $J_{Normal} = m(5 + 5) = 10m$

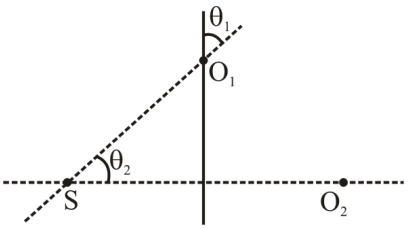
For plank, $J_{friction} = \Delta P$

$-\mu J_{Normal} = mv - m \times 10$

$- 0.1 \times 10m = mv - 10m$

$v = 9$ m/s

4) $f_1 = f_0\left(\frac{V - V_0\cos\theta_1}{V + V_1\cos\theta_2}\right)$

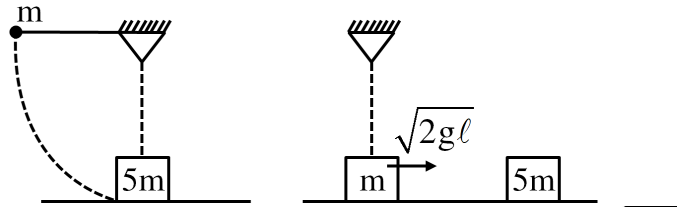


$f_2 = f_0\left(\frac{V - V_0}{V + V_1}\right)$

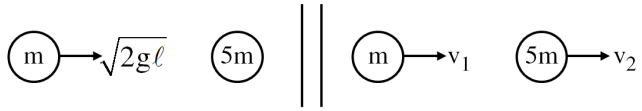
$\frac{f_1}{f_2} > 1$

$f_1 > f_2$

5) In this vertical circular motion and collision between blocks are used.



velocity of mass 'm' just before collision $v = \sqrt{2g\ell}$



$v_{CM} = \frac{m\sqrt{2g\ell} + 0}{6m} = \frac{\sqrt{2g\ell}}{6}$

$v_1 = (1 + 1)\frac{\sqrt{2g\ell}}{6} - (1)\sqrt{2g\ell}$

$$v_1 = -\frac{2}{3}\sqrt{2g\ell}$$

$$v_2 = (1+1)\frac{\sqrt{2g\ell}}{6} - (1)(0) = \frac{\sqrt{2g\ell}}{3}$$

Tension in string

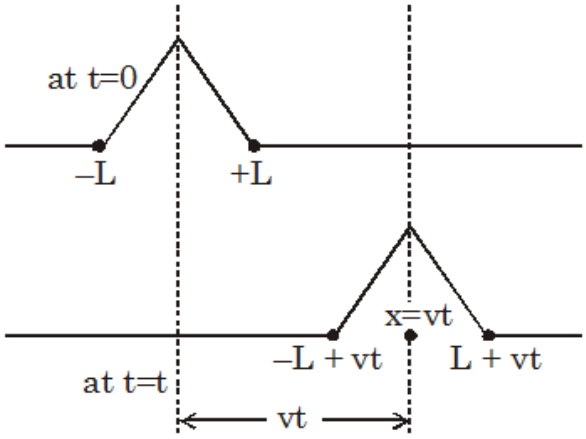
$$T - mg = \frac{mv_1^2}{\ell}$$

$$T = mg + \frac{m}{\ell}\left[\frac{4}{9} \times 2g\ell\right]$$

$$T = \frac{17}{9}mg$$

$$H_{max} = \frac{v_1^2}{2g} = \frac{4\ell}{9}$$

Correct options (1, 3, 4)



6)

$$dk = \frac{1}{2}\mu dx\, V_p^2$$

$$\frac{dk}{dx} = \frac{1}{2}\mu V_p^2 \quad \left(V_p = \text{slope} \times V\right)$$

$$\frac{dT}{dx} = \mu V_p^2$$

$$P = \frac{dT}{dt} \times \frac{dk}{dx} = V.\frac{dT}{dx}$$

$$P = V\mu V_p^2$$

$$= V\mu V^2 \times (\text{slope})^2 \quad \left(V = \sqrt{\frac{T}{\mu}}\right)$$

$$= FV\left(\frac{h}{L}\right)^2$$

Power for –L + vt < x < L + vt is $Fv\left(\frac{h}{L}\right)^2$
& zero at all other points
So, Ans. (A,B,C)

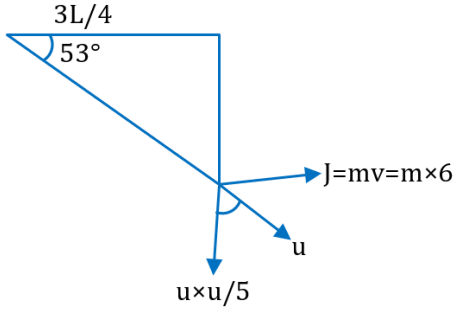
7) (a) let V_0 be the speed of sound in liquid then time taken by signal to reach from left to right

is $t_{LR} = \frac{D}{V_0 - V}$ and that from right to left is $t_{LR} = \frac{D}{V_0 - V}; \lambda_L = \lambda_0 + VT$
and $\lambda_R = \lambda_0 + VT$

$f_L = f_0 \left(\frac{V_0 + V}{V_0 + V}\right)$

$f_R = f_0 \left(\frac{V_0 - V}{V_0 - V}\right)$

8) $\tan 53 = \frac{L}{3L} \times 4 \Rightarrow \frac{4}{3}$



$2mv \Rightarrow mu \sin 53$

$2mv = m \times 15 \times \frac{4}{5}$

v = 6

9)

Velocity of B after collision will be

$V = \frac{3}{2}\frac{2mv_0}{3m} = v^0$

Tension just after the collision is

$T = \frac{mv_0^2}{\ell} + mg = 40$

$\Rightarrow v_0^2 = 30 \times 0.3$

$\Rightarrow v_0 = 3$ m/s

10)

By conservation of momentum

$-200 \times v_1 + 2 \times v_2 = 0$

$100\ v_1 = v_2$ ... (1)

$v_1 = v_2 / 100$

$\left(\frac{v_2}{100}\right) \cdot t = 8$

$t = \frac{800}{v_2}$

$v_2 \times t = x$

$x = v_2 \times \frac{800}{v_2}$

x = 800 m

11) Velocity of wave $v = \sqrt{\frac{T}{\mu}}$

$$\frac{dx}{dt} = \sqrt{\frac{Kt}{\mu}}$$

$$dx = \sqrt{\frac{K}{\mu}} \cdot t^{1/2} dt$$

On integration

$$L = \sqrt{\frac{K}{\mu}} \cdot t^{3/2} \cdot \frac{2}{3}$$

$$2 = \sqrt{\frac{K \times 10^{-2}}{2 \times 10^{-3}}} \times (L)^{3/2} \times 2/3$$

$$9 = \frac{K}{2} \times 10$$

$$\boxed{K = 1.80}$$

12) Velocity of wind in the direction of motion of A = 20 cos 60° = 10 m/s

$$f_A = \frac{340 - 10}{340 - 10 + 10} \times 85 = \frac{330}{4} Hz$$

$$f_B = \frac{340 + 10}{340 + 10 - 10} \times 85 = \frac{350}{4} Hz$$

$f_{beat} = f_B - f_A = 5$ Hz Ans.

13) $f \propto \frac{1}{\ell_1} \Rightarrow f = \frac{k}{\ell_1}$ .... (1)

($\ell_1 \Rightarrow$ initial length of pipe)

$$\left(\frac{V}{V - V_T}\right) f = \frac{k}{\ell_2}$$

{V_T = Speed of tuning fork, ℓ_2 = new length of pipe} ...(2)

(1) ÷ (2)

$$\frac{V - V_T}{V} = \frac{\ell_2}{\ell_1}$$

$$\frac{\ell_2}{\ell_1} - 1 = \frac{V - V_T}{V} - 1$$

$$\frac{\ell_2 - \ell_1}{\ell_1} = \frac{-V_T}{V}$$

$$\frac{\ell_2 - \ell_1}{\ell_1} \times 100 = \frac{-2}{320} \times 100 = -0.625$$

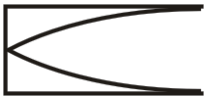
Therefore smallest value of percentage change required in the length of pipe is 0.625

Answer is (0.62 to 0.63)

14) Conceptual

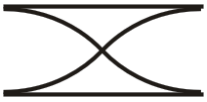
15) **New Ans. (A) by YA sir**

16) For (A) : Sound wave is longitudinal wave



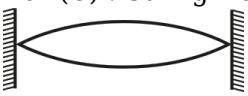
$\frac{\lambda_F}{4} = L \Rightarrow \lambda_F = 4L$

For (B) : Sound wave is longitudinal wave



$\frac{\lambda_F}{2} = L \Rightarrow \lambda_F = 2L$

For (C) : String wave is transverse



$\frac{\lambda_F}{2} = L \Rightarrow \lambda_F = 2L$

For (D) :



$\lambda_F = L \Rightarrow$ so (q & r)

PART 2 : CHEMISTRY

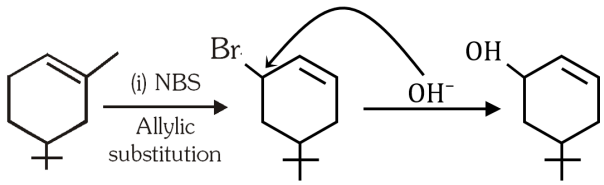
17) 1° R- X gives S_N2 reaction fastest and 3° R -X gives S_N1 reaction fastest.

18)

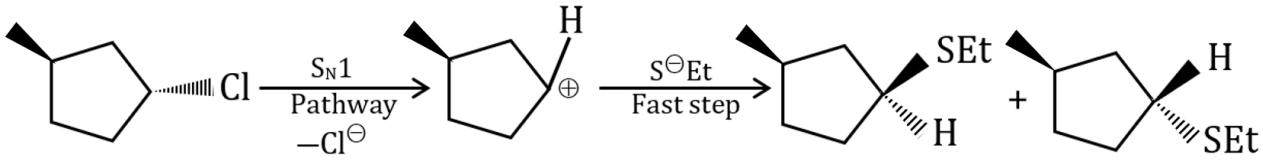
In the presence of $-\overset{\oplus}{N}R_3$, –F and bulky base reaction take via Hoffmann mechanism.

19) Rate of dehydration α stability of first carbocation

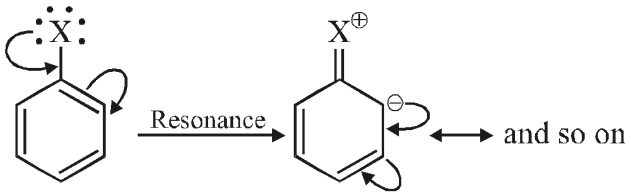
20)



21)



22) HBr in presence of H_2O_2 show anti-Makovnikov' s addition, others cause free radical substitution.



23) Arylhalide

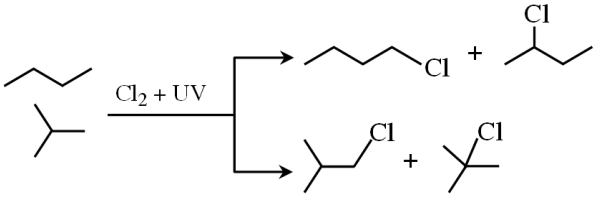
* Due to resonance partial double bond character is present in C — X bond, so it is difficult to cleavage of C — X bond

* halogen is participate in resonance because carbon of benzene is sp^2 hybridised

24) i, ii, iv, vi, viii

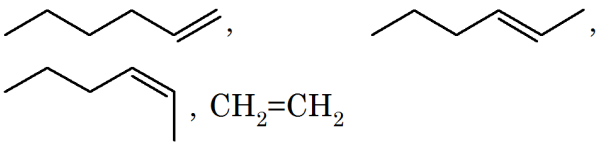
25) CH_3-Br, O_2N-CH_2Br, $OHC-CH_2-Br$ and $CH_2=CH-CH_2-Br$ are more reactive than ethyl bromide for S_N2 reaction.

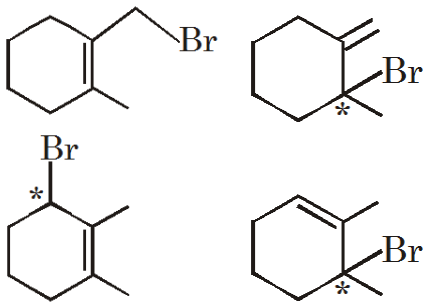
26) **New Solution :**



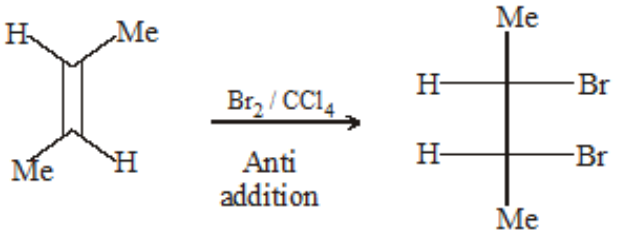
Old Solution :

All are correct (Moderate)

27) 

28) 

$\frac{7}{2}$ = 3.50



trans-2-butene meso compound

Possible stereo isomer(s) $CH_3-\overset{*}{C}H-\overset{*}{C}H-CH_3$ (Br, Br)

(n = 2)

29) Total stereo(s) = $2^{n-1} + 2^{\frac{n+1}{2}-1} = 3$

30) **Ans. :**

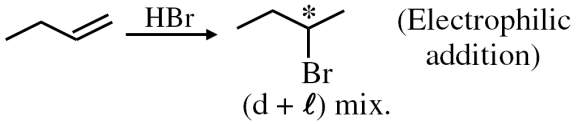
P → 3, 4, 5

Q → 3, 4, 5

R → 1, 4

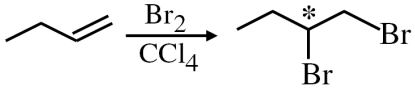
S → 2, 4

Solution :



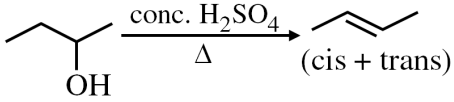
(Electrophilic addition)

(d + ℓ) mix.

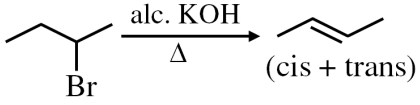


(Electrophilic addition)

(d + ℓ) mix.

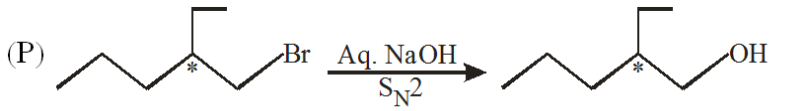


(cis + trans)

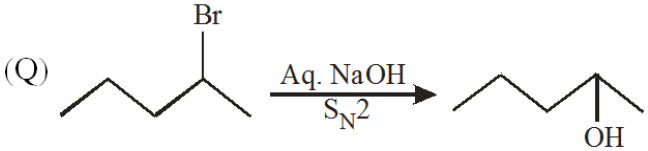


(cis + trans)

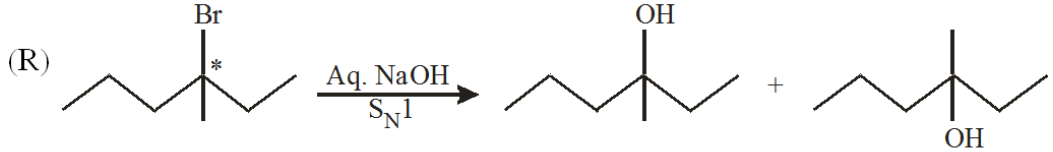
31)



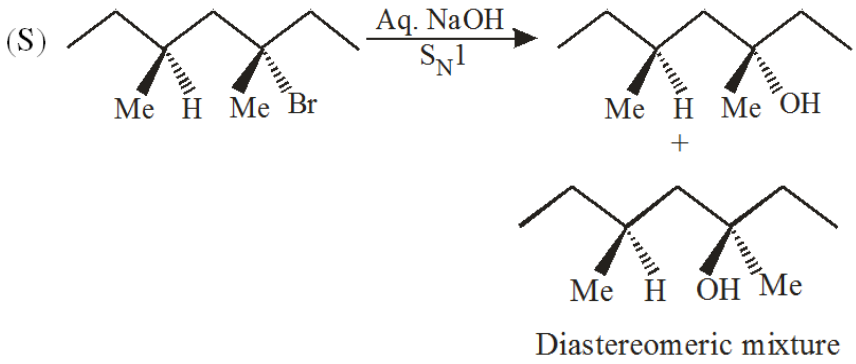
Retention of configuration



Inversion of configuration



Mixture of enantiomers



Diastereomeric mixture

32)

(A) S, T
(B) Q T S
(C) R
(D) R

PART 3 : MATHEMATICS

33) $$I = \int \frac{dx}{(x+4)^{\frac{8}{7}}(x-3)^{\frac{6}{7}}} = \int \frac{dx}{\left(\frac{x+4}{x-3}\right)^{\frac{8}{7}}(x-3)^2}$$

Let $\frac{x+4}{x-3} = t \Rightarrow \frac{dx}{(x-3)^2} = -\frac{1}{7}dt$

$$\Rightarrow I = -\frac{1}{7}\int \frac{dt}{t^{8/7}} = -\frac{1}{7}\int t^{-8/7}dt$$

$$= t^{-1/7} + C = +\left(\frac{x+4}{x-3}\right)^{-1/7} + C = \left(\frac{x-3}{x+4}\right)^{1/7} + C$$

34) $$I = \int e^x\left(\tan^{-1}x + \frac{2x}{(1+x^2)^2}\right)dx$$

$$= \int e^x\tan^{-1}x\,dx + \int e^x\frac{2x}{(1+x^2)^2}dx$$

$$= e^x\tan^{-1}x - \int \frac{e^x}{1+x^2}dx + 2\int \frac{xe^x}{(1+x^2)^2}dx$$

$$= e^x\tan^{-1}x - \left[\frac{e^x}{1+x^2} - \int \frac{-e^x.2xdx}{(1+x^2)^2}\right] + 2\int \frac{xe^x}{(1+x^2)^2}dx$$

$$= e^x\left(\tan^{-1}x - \frac{1}{1+x^2}\right) + C$$

f(0) = -1

C = 0

$$f(1) = e\left(\frac{\pi}{4} - \frac{1}{2}\right)$$

35) $I = \int_{-\pi/2}^{\pi/2} (\sin^2 x - \sin^4 x)$ (sin x – cos x)dx

$= I_1 + I_2$, where

$I_1 = \int_{-\pi/2}^{\pi/2} (\sin^3 x - \sin^5 x)dx = 0$

[$\because f(-x) = -f(x)$]

$I_2 = \int_{-\pi/2}^{\pi/2} (\sin^4 x - \sin^2 x)\cos x\,dx$

$= 2\int_0^1 (t^4 - t^2)dt$ [$\because f(-x) = f(x)$ and sin x = t]

$= 2\left[\frac{1}{5} - \frac{1}{3}\right] = -\frac{4}{15}$

I = 0 – 4/15 = –4/15

36)

In the integral J, substitute x + 1 = t

$\Rightarrow \quad dx = dt \text{ and } x^2 + 2x = (t^2 - 1)$

Now $J = \int_1^e \frac{e^{\frac{t^2-2}{2}}}{t}dt$ and $K = \int_1^e t \ln t\, e^{\frac{t^2-2}{2}}dt$

$= \int_1^e e^{\frac{t^2-2}{2}}\left(\frac{1}{t} + t\ln t\right)dt$

Hence (J + K)

$= \left(e^{\frac{t^2-2}{2}}\ln t\right)_{t=1}^{t=e} = e^{\frac{e^2-2}{2}} = \left(\sqrt{e}\right)^{e^2-2}$

37) $\therefore \int \frac{\frac{x^2-1}{x^2}}{\left(\frac{1}{x^4} + \frac{1}{x^2}\right)\sqrt{x^8 + x^6 + x^4}}dx = \sec^{-1} f(x) + C$

$\Rightarrow \int \frac{\left(1 - \frac{1}{x^2}\right)}{\frac{1}{x^3}\left(x + \frac{1}{x}\right)x^3\sqrt{x^2 + 1 + \frac{1}{x^2}}}dx = \sec^{-1} f(x) + C$

$\Rightarrow \int \frac{\left(1 - \frac{1}{x^2}\right)dx}{\left(x + \frac{1}{x}\right)\sqrt{\left(x + \frac{1}{x}\right)^2 - 1}} = \sec^{-1} f(x) + C$

$\Rightarrow \sec^{-1}\left(x + \frac{1}{x}\right) + C = \sec^{-1}(f(x)) + C$

$\Rightarrow f'(x) = \frac{(x+1)(x-1)}{x^2} = 0$

$\Rightarrow x = \pm 1$

Maximum occur at x = –1 and minimum occur at x = 1

38) $I_m = \int_0^{m\pi} x\,|\sin x|\, e^{\cos 4x}dx$ .... (1)

by using the property

$$\int_0^a f(x)dx = \int_0^a f(a-x)\,dx$$

$$I_m = \int_0^{m\pi} (m\pi - x)\,|\sin x|\, e^{\cos 4x} dx \quad \text{....(2)}$$

adding eq. (1) & eq. (2) we get

$$2I_m = m\pi \int_0^{m\pi} |\sin x|\, e^{\cos 4x} dx$$

by using periodic property

$$2I_m = m^2\pi \int_0^{\pi} |\sin x|\, e^{\cos 4x} dx$$

$$2I_m = 2m^2\pi \int_0^{\pi/2} \sin x\, e^{\cos 4x} dx$$

$$\Rightarrow I_m = m^2\pi \int_0^{\pi/2} \sin x\, e^{\cos 4x} dx \quad \text{...(3)}$$

and

$$J_n = n\int_0^{\pi} |\cos x|\, e^{\cos 4x} dx = 2n\int_0^{\pi/2} \sin x e^{\cos 4x} dx \quad \text{.... (4)}$$

on dividing eq. (3) and (4)

$\frac{I_m}{J_n} = \frac{m^2\pi}{2n}$ now B,D is correct

39) **(A)** $(a + c - b)(a - c + b) = 4bc$

$(a + (c - b))(a - (c - b)) = 4bc$

$a^2 - (c - b)^2 = 4bc$

$a^2 - (c^2 + b^2 - 2bc) = 4bc$

$a^2 = b^2 + c^2 + 2bc$

$\Rightarrow b^2 + c^2 - a^2 = -2bc$

$\Rightarrow \frac{b^2 + c^2 - a^2}{2bc} = -1$

$\Rightarrow \cos A = -1$

which is not possible.

(B) $b^2\sin^2 C + c^2\sin^2 B = ab$

$\Rightarrow (4R^2\sin^2 B)2\sin C.\cos C + (4R^2\sin^2 C)2\sin B\cos B = ab$

$\Rightarrow 8R^2.\sin B.\sin C.[\sin B.\cos C + \sin C.\sin B]$

$= (2R\sin A)(2R\sin B)$

$\Rightarrow 2\sin C.\sin(B + C) = \sin A$

$\Rightarrow 2\sin C = 1$

$\Rightarrow \sin C = \frac{1}{2}$

$\Rightarrow \angle C = \frac{\pi}{6}$ or $\frac{5\pi}{6}$ so Δ is possible.

(C) For a = 3, b = 5, c = 7 and $C = \frac{2\pi}{3}$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{5^2 + 7^2 - 3^2}{2(5)(7)} = \frac{65}{70} < 1$$

similarly

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac} = \frac{3^2 + 7^2 - 5^2}{2(3)(7)} = \frac{33}{42} < 1$$

So in this case triangle is possible

(D) As, $\cos\left(\frac{A-C}{2}\right) = \cos\left(\frac{A+C}{2}\right)$

$$\Rightarrow \cos\left(\frac{A-C}{2}\right) - \cos\left(\frac{A+C}{2}\right) = 0$$

$$\Rightarrow 2\sin\frac{A}{2}.\sin\frac{C}{2} = 0$$

which is not possible in ΔABC

40) $G(x) = \int \frac{\sin x}{\sin x - \cos x}\,dx = \frac{1}{2}\int \frac{\sin x + \cos x + \sin x - \cos x}{\sin x - \cos x}dx$

$$= \frac{1}{2}\int \frac{\sin x + \cos x}{\sin x - \cos x}dx + \frac{1}{2}x + c = \frac{1}{2}\ell n\,|\sin x - \cos x| + \frac{1}{2}x + c$$

Now, $G\left(\frac{\pi}{3}\right) = \frac{1}{2}\ell n\left(\frac{\sqrt{3}-1}{2}\right) + \frac{\pi}{6} + c$

$$G\left(\frac{\pi}{6}\right) = \frac{1}{2}\ell n\left|\frac{\sqrt{3}-1}{2}\right| + \frac{\pi}{12} + c$$

$$\square\ G\left(\frac{\pi}{3}\right) = G\left(\frac{\pi}{6}\right) = \frac{\pi}{6} - \frac{\pi}{12} = \frac{\pi}{12}$$

$$\frac{24}{\pi}\left(G\left(\frac{\pi}{3}\right) - G\left(\frac{\pi}{6}\right)\right) = 2$$

41) $f(x) = \int \left(x^2 + 2x^4 + 3x^6\right)\left(1 + x^2 + x^4\right)^{\frac{1}{2}}dx$

$$f(x) = \int \left(x + 2x^3 + 3x^5\right)\left(x^2 + x^4 + x^6\right)^{\frac{1}{2}}dx$$

$x^2 + x^4 + x^6 = t^2$

$(x + 2x^3 + 3x^5)dx = tdt$

$$f(x) = \int t^2.dt = \frac{t^3}{3} + C$$

$$f(x) = \frac{\left(x^2 + x^4 + x^6\right)^{\frac{3}{2}}}{3} + C$$

$x = 0 \quad C = f(0) = 0$

$$f(1) = \frac{3^{\frac{3}{2}}}{3} = \sqrt{3}$$

$f^2(1) = 3$

42)

$$I_{(1,10)} = \int \underset{II}{\cos 2x} \underset{I}{\left(1-\tan^2 x\right)^{10}} dx$$

$$= \frac{\sin 2x}{2}(1-\tan^2 x)^{10}$$

$$- \int \frac{\sin 2x}{2} 10(1-\tan^2 x)^9 (-2\tan x \sec^2 x) dx$$

$$= \frac{\sin 2x}{2}(1-\tan^2 x)^{10} + 20\int \tan^2 x\left(1-\tan^2 x\right)^9 dx$$

$$= \frac{\sin 2x}{2}\left(1-\tan^2 x\right)^{10}$$

$$+20\int (1-(1-\tan^2 x))(1-\tan^2 x)^9 dx$$

$$I_{(1,10)} = \frac{\sin 2x}{2}(1-\tan^2 x)^{10} + 20I_{(0,9)} - 20I_{(0,10)} + C$$

$$20I_{(0,10)} + I_{(1,10)}$$

$$= \frac{\sin 2x}{2}(1-\tan^2 x)^{10} + 20I_{(0,9)} + C$$

$$4I_{(0,10)} + \frac{I_{(1,10)}}{5}$$

$$= \frac{\sin 2x}{10}(1-\tan^2 x)^{10} + 4I_{(0,9)} + C$$

$$A = \frac{1}{10};\ B = 4$$

43)

Substitute $x = \frac{ab}{t}$.

44)

Let $h = \frac{1}{n}$, so

$$\lim_{n\to\infty} n^2 \int_0^{1/n} x^{x+1} dx = \lim_{h\to 0^+} \frac{\int_0^h x^{x+1} dx}{h^2}$$

$$= \lim_{h\to 0^+} \frac{\frac{d}{dh}\int_0^h x^{x+1} dx}{\frac{d}{dh}h^2}$$

$$= \lim_{h\to 0^+} \frac{h^{h+1} - 0^1}{2h}$$

$$= \frac{1}{2}\lim_{h\to 0^+} h^h$$

$$= \frac{1}{2}\exp\left(\lim_{h\to 0^+} \ln(h^h)\right)$$

$$= \frac{1}{2}\exp\left(\lim_{h\to 0^+} h\ln(h)\right)$$

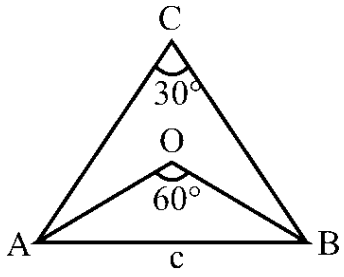
$$= \frac{1}{2}\exp\left(\lim_{h\to 0^+} \frac{\ln(h)}{1/h}\right)$$

$$= \frac{1}{2}\exp\left(\lim_{h\to 0^+} \frac{\frac{d}{dh}\ln(h)}{\frac{d}{dh}\frac{1}{h}}\right)$$

(#)

$$= \frac{1}{2}\exp\left(\lim_{h\to 0^+}\frac{1/h}{-1/h^2}\right)$$
$$= \frac{1}{2}\exp\left(\lim_{h\to 0^+} -h\right)$$
$$= \frac{1}{2}\exp(0)$$
$$= \frac{1}{2}$$

and our answer is 3. Here, the statements marked with (#) are derived using L'Hopital's rule, and the rest is algebraic manipulation.

45) 

$$1 - 2\sin\left(\frac{3A-B}{2}\right)\cos\frac{C}{2} = \cos C + 2\cos 2B$$
$$1 - \left[2\sin\left(\frac{3A-B}{2}\right)\sin\left(\frac{B+A}{2}\right)\right] = \cos C + 2\cos 2B$$

Using sum and product formulas and replace

$$\sin A = \frac{a}{2R},\ \sin B = \frac{b}{2R},\ \sin C = \frac{c}{2R}$$

We get $2b^2 - ab - a^2 = 0$

$\Rightarrow$ a = b or a = –2b (reject)

$\Rightarrow$ $\angle A = \angle B = \frac{5\pi}{12},\ C = \frac{\pi}{6}$

Put $R_1 = \frac{c}{2\sin C}$

& $R_2 = \frac{c}{2\sin 2C}$

46) (P) $\because$ $2B = A + C \Rightarrow B = \frac{\pi}{3}$ and $A + C = \frac{2\pi}{3}$

$\because$ $b^2 = ac$

$\Rightarrow \sin^2 B = \sin A.\sin C$

$\Rightarrow \sin A \sin C = \frac{3}{4}$

$\Rightarrow \cos(A - C) - \cos(A + C) = \frac{3}{2}$ $\because$ $A + C = \frac{2\pi}{3}$

$\Rightarrow \cos(A - C) = 1$

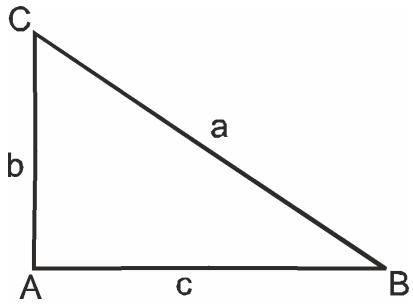
$\Rightarrow A = C = \frac{\pi}{3} = B$

$\Rightarrow a = b = c$

$\Rightarrow \frac{a^2(a+b+c)}{3abc} = 1$

(Q) $\because$ $a^2 = b^2 + c^2$ and $2R = a$

$\Rightarrow \frac{a^2+b^2+c^2}{R^2} = \frac{2a^2}{R^2} = 8$



(R) $\because \Delta = \frac{1}{2}bc\sin A \Rightarrow \Delta = \frac{1}{2}.9.\sin A = \frac{9}{2} \times \frac{a}{2R}$ $\because a = 2$

$\Rightarrow 2R\Delta = 9$

(S) $\because$ $a = 5$, $b = 3$ and $c = 7$

and because we know that

$b\cos C + c\cos B = a$

$\Rightarrow 3\cos C + 7\cos B = 5$

47) (P)
$$I_1 = \int_{-\pi/2}^{\pi/2} \cos^{-1}(\sin x)\,dx = \int_{-\pi/2}^{\pi/2} \left(\frac{\pi}{2} - \sin^{-1}(\sin x)\right) dx$$
$$= \int_{-\pi/2}^{\pi/2} \left(\frac{\pi}{2} - x\right) dx = \frac{\pi}{2}\int_{-\pi/2}^{\pi/2} dx = \frac{\pi}{2}(\pi) = \frac{\pi^2}{2}$$
$$\Rightarrow \frac{8I_1}{\pi^2} = \frac{8}{\pi^2}.\frac{\pi^2}{2} = 4$$

(Q)
$$I_2 = \int_0^{\pi} \tan^{-1}(\cos x)\,dx = \int_0^{\pi} \tan^{-1}(\cos(\pi - x))\,dx$$
$$= \int_0^{\pi} \tan^{-1}(-\cos x)\,dx$$
$$= -\int_0^{\pi} \tan^{-1}(\cos x)\,dx$$
$$\Rightarrow I_2 = 0$$

(R)
$$I_3 = \int_0^{\pi/2} x^2 \sin 2x\,dx$$
Let $2x = t \Rightarrow dx = \frac{dt}{2}$
$$= \frac{1}{2}\int_0^{\pi} \frac{t^2}{4}\sin t\,dt = \frac{1}{8}\int_0^{\pi} \underbrace{t^2}_{I}\,\underbrace{\sin t}_{II}\,dt$$
$$8I_3 = -t^2\cos t\Big]_0^{\pi} + 2\int_0^{\pi} t\cos t\,dt$$
$$= -\left[-\pi^2\right] + 2\left[t\sin t\Big]_0^{\pi} - \int_0^{\pi}\sin t\,dt\right]$$
$$= \pi^2 + 2[(-1) - 1] = \pi^2 - 4 \Rightarrow 4 = (\pi^2 - 8I_3)$$

Hence, $\frac{3}{4}(\pi^2 - 8I_3) = \frac{3}{4} \times 4 = 3$

(S) $I_4 = \int_{\pi}^{2\pi} x^{[\sin x]}\,dx$

in $(\pi, 2\pi)$, $[\sin x] = -1$

$\Rightarrow \int_{\pi}^{2\pi} \frac{1}{x}\,dx = [\ell nx]_{\pi}^{2\pi}$

$= \ell n2\pi - \ell n\pi = \ell n2$

$\Rightarrow e^{I_4} = e^{\ell n2} = 2$

48) (P) $\int \frac{2^x}{\sqrt{1-4^x}}dx = k\sin^{-1} f(x) + C$

let $2^x = t$

$\frac{1}{\ell n2}\int \frac{dt}{\sqrt{1-t^2}} = \frac{1}{\ell n2}\sin^{-1}(2^x) + C$

$\Rightarrow k = \frac{1}{\ell n2} > 1$

(Q) $\int \frac{x^{5/2}}{x^{7/2}+x^6}dx = \int \frac{dx}{x\left(1+x^{5/2}\right)}$

$= \int \frac{dx}{x} - \int \frac{x^{3/2}}{1+x^{5/2}}dx$

$= \ell nx - \frac{2}{5}\ell n\left(1+x^{5/2}\right) + C$

$= \frac{2}{5}\ell n\left(\frac{x^{5/2}}{1+x^{5/2}}\right) + C$

$a = \frac{2}{5}, k = \frac{5}{2}$

(R) $\int \frac{x^2+1}{x}dx - \int \frac{2x}{x^2+1}dx$

$= \frac{x^2}{2} + \ell nx - \ell n\left(1+x^2\right) + C$

$k = 2$

(S) $\int \frac{\sec^2\frac{x}{2}dx}{9-\tan^2\frac{x}{2}} = \int \frac{2dt}{9-t^2}$

$= \frac{2}{3}\tan^{-1}\left(\frac{1}{3}\tan\frac{x}{2}\right) + C$

$k = \frac{2}{3}, m = \frac{1}{3}, \frac{k}{m} = 2$